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QuiZForge: AI-Powered Quiz & Exam Preparation Platform

Software Requirement Engineering

Sec: B

Project submitted

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1. PROBLEM DOMAIN

1.1 Background to the Problem

A big problem in Bangladesh education is that from secondary school to the university level, students do not have access to quality learning tools that are based on best practice. Traditional teaching is mainly 'lecture' and 'test manually', and students have little chance to prepare for tests. Teachers, meanwhile, are spending too much time developing quizzes and tests, which they could be better using to teach students.

Even though the whole world went digital with the outbreak of COVID-19, Bangladesh has been far behind in the intelligent use of educational technology. The majority of students use printed instructions or static PDFs without providing any feedback. The consequences are low performance, test anxiety, and widening the achievement gap.

UNESCO and the Bangladesh Bureau of Educational Information and Statistics (BANBEIS) state that there are more than 40 million students in the educational institutions of Bangladesh. Less than 5% have access to any AI-assisted or adaptive learning tool. The opportunity is huge, and the problem is societal.

1.2 Solution to the Problem

QuizForge is a website that helps teachers and students prepare for quizzes and exams. It does this by letting teachers make quizzes from the things they have already written, like notes or PDF files. Students can then use these quizzes to practice and get feedback.

This is an idea because:

- It is made for two important groups of people: teachers who make the questions and students who use them to learn
- It uses computer programs that can understand what people are saying to make the hard work of making quizzes easier
- It works in two languages: Bengali and English. More people in Bangladesh can use it
- You can use it from any device that is connected to the internet. You do not need to put anything on your device to use it

This is something that can actually be done because there are companies that can help us with the computer part. It is easy to have a website in Bangladesh, and many other websites, like Quizlet and Kahoot, have already shown that this is a good way to do things. QuizForge is a website that helps with quiz and exam preparation.

1.3 Existing Software Solutions

Several global platforms partially addressed this problem, but none fully serve the Bangladeshi context:

Platform	Strengths	Weakness
Quizlet	Flashcard-based learning, global user base	No AI quiz generation from PDFs, no Bengali support
Google Forms	Simple quiz creation, free	No AI generation, no analytics, no student tracking
Coursera/edX	Full courses available	Not focused on school/university exam prep, expensive
Kahoot!	Engaging game-based quizzes	Requires live session, not self-paced, no local curriculum
QuizForge	AI generation, Bengali+English, local curriculum, analytics	New platform (our solution)

QuizForge fills the critical gap by combining AI-powered content generation, local language support, student analytics, and curriculum alignment — all in one affordable, web-based platform.

2. SOLUTION DESCRIPTION

QuizForge is a website that schools and teachers in Bangladesh can use. It is also for people who want to study on their own. There are three types of users on QuizForge: the Admin, the Teacher, and the Student. Teachers put study materials on QuizForge, set up quizzes, and then the computer system looks at the materials. Makes questions. QuizForge is a tool for students because they can take quizzes and find out how they did right away. QuizForge is a place for students to learn and for teachers to help them. The Admin, the Teacher, and the Student all use QuizForge to do things. QuizForge is a website that people in Bangladesh can use to study and learn.

2.1 System Features

Feature 1: User Authentication & Role Management

- FR-01: Users can register as a Teacher or a Student with an email and a password
- FR-02: Admin can manage all users, courses, and platform settings
- FR-03: Users can reset their passwords via email OTP
- FR-04: Session management with JWT-based authentication
- QA: Security, passwords hashed with bcrypt; sessions expire after inactivity

Feature 2: AI-Powered Quiz Generation

- FR-05: Teachers can upload PDF, DOCX, or plain text files as source material
- FR-06: The AI engine (NLP-based) extracts key concepts and generates MCQ, True/False, and short-answer questions
- FR-07: Teachers can review, edit, and approve generated questions before publishing
- FR-08: Supports question generation in English and Bengali
- QA: Accuracy, AI-generated questions reviewed before publishing; fallback manual entry always available

Feature 3: Quiz Management

- FR-09: Teachers can set the quiz title, time limit, difficulty level, and passing marks
- FR-10: Teachers can assign quizzes to specific student groups or make them public
- FR-11: Quiz questions can be shuffled to prevent cheating
- FR-12: Teachers can set quiz availability windows (start/end date)
- QA: Reliability — quizzes auto-save student progress; no loss on accidental refresh

Feature 4: Student Quiz Taking

- FR-13: Students can browse available quizzes by subject or teacher
- FR-14: Students receive a countdown timer during quiz sessions
- FR-15: Students receive immediate feedback after each question (optional) or at quiz end
- FR-16: Students can review incorrect answers with correct explanations
- QA: Usability — mobile-responsive design; accessible on phones and tablets

Feature 5: Performance Analytics Dashboard

- FR-17: Students can view personal performance history: score trends, weak topics, and time spent
- FR-18: Teachers can view class-level analytics: average scores, question difficulty stats
- FR-19: Exportable reports in PDF/CSV for teachers
- QA: Performance — dashboard loads within 2 seconds; handles up to 500 concurrent users

Feature 6: Subscription & Payment

- FR-20: Free tier with limited quiz generation (5 quizzes/month)
- FR-21: Pro Teacher plan with unlimited generation and advanced analytics
- FR-22: Institution plan for schools and universities with multi-teacher support
- FR-23: Payment via bKash, Nagad, and credit/debit card
- QA: Availability — 99.5% uptime SLA; payment gateway with SSL encryption

2.2 UML Diagrams

○ Use Case

The Use Case Diagram below identifies the primary actors and their interactions with the QuizForge system:

Actors: Admin, Teacher, Student, AI Engine (system actor)

Teacher Use Cases:

- Register/Login
- Upload Study Material
- Generate Quiz (via AI Engine)
- Review and Edit Questions
- Assign Quiz to Students
- View Analytics Dashboard

Student Use Cases:

- Register/Login
- Browse Available Quizzes
- Take Quiz
- View Results & Feedback
- Track Personal Performance

Admin Use Cases:

- Manage Users (CRUD)
- Manage Subscriptions
- View Platform Reports

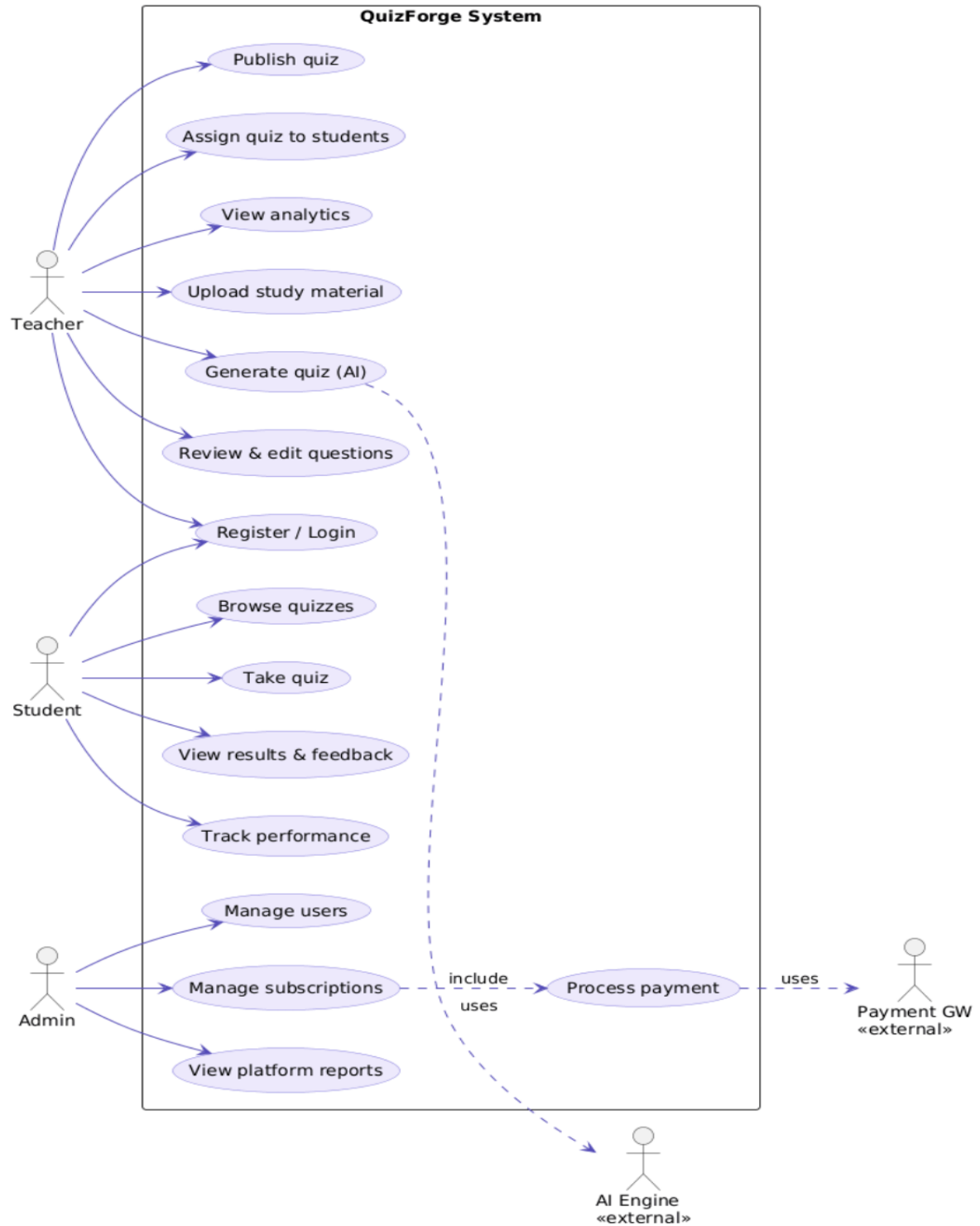


Figure 1: Use Case Diagram.

○ Activity Diagram

The Activity Diagram below describes the process flow when a teacher generates a quiz using the AI engine:

Flow Steps:

1. Teacher logs in to the platform
2. Teacher navigates to the 'Create Quiz' section
3. The teacher uploads a PDF or enters text content
4. The system sends content to the AI Engine.
5. AI Engine extracts key concepts and generates question candidates
6. Decision: Are the generated questions satisfactory?
7. Yes → Teacher reviews, edits, and publishes the quiz
8. No → Teacher manually adds/edits questions and publishes
9. The system stores the quiz in the database and notifies the assigned students
10. Students receive quiz availability notification.



Figure 2: Activity Diagram.

○ Class Diagram

The Class Diagram below defines the main classes and their relationships in the QuizForge system:

Main Classes:

- User (id, name, email, passwordHash, role, createdAt) — base class
- Teacher extends User (teacherId, institution, subscriptionPlan)
- Student extends User (studentId, enrolledCourses, performanceHistory)
- Quiz (quizId, title, createdBy: Teacher, questions: Question, timeLimit, difficulty, status)
- Question (questionId, text, type: MCQ/TF/Short, options, correctAnswer, explanation, language)
- QuizAttempt (attemptId, student: Student, quiz: Quiz, answers, score, startedAt, completedAt)
- StudyMaterial (materialId, uploadedBy: Teacher, fileUrl, processedStatus, extractedText)
- Subscription (subscriptionId, user: User, plan, startDate, endDate, paymentStatus)

Relationships:

- Teacher (1) --creates--> (many) Quiz
- Quiz (1) --contains--> (many) Question
- Student (1) --attempts--> (many) QuizAttempt
- QuizAttempt (many) --references--> (1) Quiz
- Teacher (1) --uploads--> (many) StudyMaterial
- StudyMaterial (1) --generates--> (many) Question

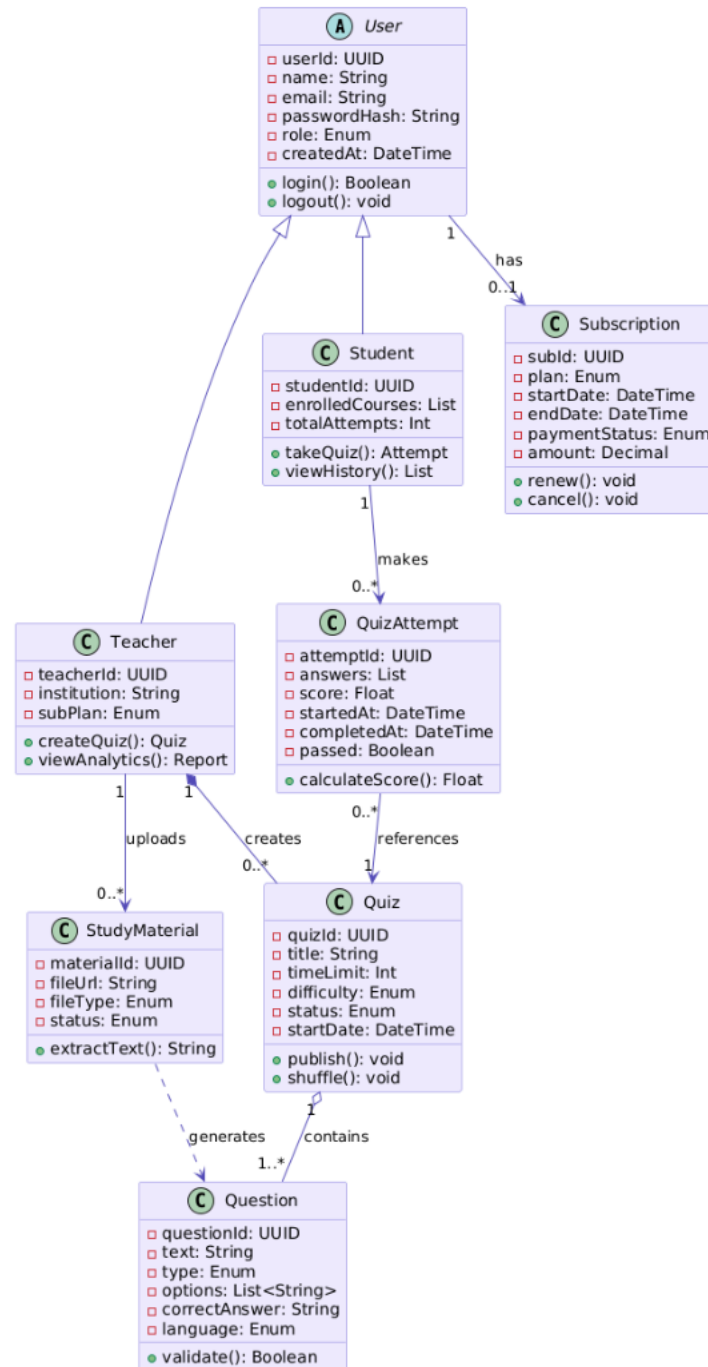


Figure 3: Class Diagram.

3. Social Impact

QuizForge has the potential to create significant positive social impact across multiple dimensions of Bangladeshi society:

3.1 Improved Educational Outcomes

Picture a student in Sylhet the night before her HSC exam. She has her textbook, a few handwritten notes, and no way to know whether she actually understands the material or just thinks she does. No practice tests. No feedback. No way to find out until the exam itself.

This is the reality for the overwhelming majority of students in Bangladesh. QuizForge changes that dynamic by giving students a way to test themselves, identify their weak spots, and keep practicing — on their own schedule, from their own phone. Students who practice with immediate feedback don't just score better. They walk into exams with something harder to teach: confidence.

3.2 Empowering Teachers

Teachers in Bangladesh are overworked in a way that rarely gets acknowledged. Many spend five to eight hours every week doing nothing but writing quiz questions — time that could have gone toward an extra explanation, a struggling student who needed attention, or simply recovering enough to teach well the next day.

QuizForge gives that time back. What used to take an entire evening now takes thirty minutes. That might sound like a small thing, but for a teacher managing forty students across multiple subjects, it is not small at all. One well-rested, well-prepared teacher can change the trajectory of hundreds of students. QuizForge just clears the path.

3.3 Democratizing Access to Quality Education

In Bangladesh, exam preparation has quietly become a privilege. If your family can afford a coaching center, you get structured practice, mock tests, and guidance. If they cannot, you make do. That divide has nothing to do with how intelligent or hardworking a student is — it is purely economic.

QuizForge refuses to accept that as a fixed reality. The free tier means any student with a phone and an internet connection can access the same quality of practice material as someone paying for private coaching. Bengali language support means students are not penalized for not being fluent in English. These are not just features — they are a deliberate choice about who this platform is built for.

3.4 Digital Literacy and Skill Development

When a student uses QuizForge regularly, they are learning more than biology or accounting. They are learning how to interact with digital tools, how to read performance data, and how to self-direct their own learning. These are exactly the skills that employers in Bangladesh's growing tech and service sectors are looking for — and they are skills that no textbook explicitly teaches. Bangladesh has set an ambitious goal of becoming a knowledge-based economy by 2041. That goal lives or dies on whether the current student generation is ready for it. Platforms like QuizForge are one small but concrete step in that direction.

3.5 Environmental Impact

Using things instead of paper-based tests and printed study guides, and physical assessment sheets can really cut down on the amount of paper we use. If one school starts using QuizForge, they can save a lot of paper every year. We are talking about thousands of sheets of paper that QuizForge can help save each year. This is a deal for QuizForge and the environment. QuizForge is a tool for schools to use, and it can make a big difference in paper usage.

4. Development Plan with Project Schedule

4.1 Software Development Lifecycle (SDLC) — Agile Methodology

QuizForge will be developed using the Agile (Scrum) methodology, organized into 4 phases over 12 months. Each phase consists of 2-week sprints with defined deliverables.

4.2 Development Phases

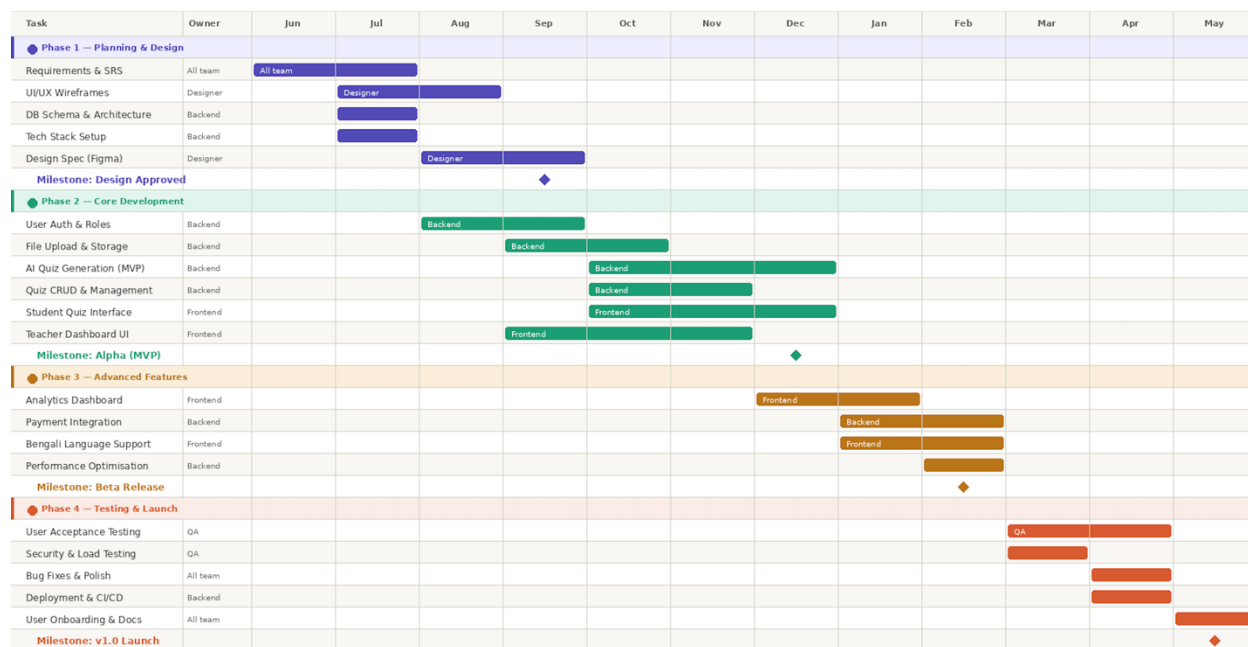
Phase	Activities	Duration in Months	Deliverables
Phase 1: Planning & Design	Requirements gathering, SRS, system design, UI/UX wireframes, database schema	1- 3	SRS Document, Wireframes, DB Schema
Phase 2: Core Development	User auth, file upload, AI integration, quiz CRUD, basic student interface	3 – 6	Working MVP (Alpha)
Phase 3: Advanced Features	Analytics dashboard, payment integration, Bengali support, performance optimization	7 – 9	Beta Release
Phase 4: Testing & Launch	UAT, security testing, bug fixes, deployment, user onboarding	10 – 12	v1.0 Production Launch

4.3 Technology Stack

Layer	Technology	Reason
Frontend	React.js + Tailwind CSS	Fast, responsive, component-based UI
Backend	Node.js + Express.js	Scalable REST API, large ecosystem
Database	PostgreSQL + Redis (cache)	Relational data + fast session/cache
File Storage	AWS S3 / Cloudflare R2	Secure PDF/document storage
Hosting	AWS EC2 / DigitalOcean	Reliable, scalable cloud hosting
AI Engine	OpenAI GPT API / HuggingFace	NLP for quiz generation
Payment	bKash PGW + SSLCommerz	Local Bangladesh payment support

4.4 Project Schedule

The detailed project schedule has been prepared using a Gantt chart in Jira / ProjectLibre / Trello. The schedule includes all SDLC phases, sprint breakdowns, milestones, and task assignments.



4.5 Team Roles

Role	Responsibilities
Project Lead / Backend Developer	Architecture, API development, AI integration, project coordination
Frontend Developer	React UI, responsive design, quiz-taking interface
UI/UX Designer	Wireframes, design system, user experience testing
QA / Documentation	Testing, SRS documentation, report writing

5. Marketing Plan

QuizForge's marketing strategy is designed to build awareness, acquire users, and retain them through a combination of digital, community, and institutional approaches.

5.1 Target Market

- Primary: University and college students in Bangladesh (age 18-26) (approximately 4 million)
- Secondary: School students (SSC/HSC level) (approximately 10 million)
- Tertiary: Teachers and educational institutions across Bangladesh

5.2 Short – Term Marketing Plan (0-6 Months)

Goal: Achieve 5,000 registered users and 50 teacher accounts within 6 months of launch.

Strategy	Action	Channel
Campus Launch Campaign	Free workshops at AIUB, BUET, DU, NSU introducing QuizForge	In-person + Social Media
Social Media Marketing	Daily posts, study tips, demo videos on Facebook, Instagram, TikTok	Facebook, Instagram, TikTok
Referral Program	Existing users get 1 month free Pro for every 3 referrals	In-app + Email
Influencer Outreach	Partner with popular Bangladeshi education YouTubers/TikTokers	YouTube, TikTok
Free Teacher Onboarding	Offer 3 months free Pro plan to first 100 teachers who sign up	Email, Direct Outreach

5.3 Long-Term Marketing Plan (6-24 Months)

Goal: 100,000 active users, partnerships with 50+ institutions, and BDT 1 Crore annual revenue.

Strategy	Actions	Timeline
Institutional Partnerships	Sign MoUs with schools and coaching centers for bulk subscriptions	Month 6-12
SEO & Content Marketing	Blog posts, study guides, ranking on Google Bangladesh for exam prep queries	Month 3 onwards
Google/Facebook Ads	Paid campaigns targeting students during HSC/SSC exam seasons	Month 6-18
Strategy	Actions	Timeline
Mobile App Launch	Extend QuizForge to Android/iOS for broader reach	Month 12-18
Government/NGO Tie-ups	Partner with a2i (Access to Information) program for rural school access	Month 12-24

5.4 Continuous Marketing Plan (Ongoing)

- Email newsletters with study tips, new feature announcements, and performance insights
- Annual 'QuizForge Academic Challenge' — national online quiz competition for students
- Loyalty program: reward active students with badge certificates and leaderboard recognition
- Community building: active Facebook group and Discord server for students and teachers
- Regular press releases and features in EdTech Bangladesh media outlets.

6. Cost and Profit Analysis

6.1 Development Cost

Cost Item	Type	Estimated Cost (BDT)
Lead Developer (12 months)	Human Resource	3,60,000
Frontend Developer (12 months)	Human Resource	2,40,000
UI/UX Designer (6 months)	Human Resource	90,000
QA Engineer (6 months)	Human Resource	72,000
AWS Hosting (1 year)	Infrastructure	60,000
OpenAI API Credits (1 year)	Infrastructure	1,20,000

Domain & SSL	Infrastructure	5,000
bKash/SSLCommerz Integration	Integration	15,000
Testing & QA Tools	Tools	20,000
Contingency (10%)	Buffer	98,200
TOTAL DEVELOPMENT COST		10,80,200

6.2 Marketing Cost

Marketing Activity	Period	Estimated Cost (BDT)
Social Media Ads (Facebook/Instagram)	Year 1	60,000
Influencer Partnerships	Year 1	40,000
Campus Events & Workshops	Year 1	50,000
SEO / Content Creation	Year 1	30,000
Print Materials (Brochures, Posters)	Year 1	15,000
TOTAL MARKETING COST (Year 1)		1,95,000

6.3 Revenue Model

Plan	Price (BDT/month)	Features	Target Users
Free	0	5 AI quizzes/month, basic analytics, max 30 students	Individual students, trial teachers
Pro Teacher	499	Unlimited AI quizzes, full analytics, up to 200 students, export reports	Individual teachers, tutors
Institution	4,999	Multi-teacher, unlimited everything, custom branding, priority support	Schools, coaching centers, universities

6.4 Revenue Projections

Year	Free Users	Pro Teachers	Institutions	Monthly Revenue (BDT)	Annual Revenue (BDT)
Year 1	5,000	100	5	74,900	8,98,800
Year 2	25,000	500	20	3,49,800	41,97,600
Year 3	80,000	1,500	60	10,24,500	1,22,94,000

6.5 Profit Analysis

Year	Revenue (BDT)	Total Cost (BDT)	Net Profit (BDT)	Profit Margin
Year 1	8,98,800	12,75,200	3,76,400	42% (Investment Phase)
Year 2	41,97,600	18,00,000	23,97,600	57% (Break-even & Growth)
Year 3	1,22,94,000	25,00,000	97,94,000	80% (High Profitability)

The analysis shows QuizForge breaks even in Year 2 and becomes highly profitable in Year 3 with a projected net profit of BDT ~98 Lakhs. This makes QuizForge an attractive investment opportunity with strong ROI for investors.

7. Reference

1. Bangladesh Bureau of Educational Information and Statistics (BANBEIS). (2023). Bangladesh Education Statistics 2023. Ministry of Education, Bangladesh.
2. UNESCO. (2022). Digital Transformation of Education in Bangladesh. United Nations Educational, Scientific and Cultural Organization.
3. Quizlet Inc. (2024). Quizlet for Teachers — Product Overview. <https://quizlet.com/teachers>
4. Kahoot! AS. (2024). Kahoot! Platform Overview. <https://kahoot.com>
5. OpenAI. (2024). GPT-4 API Documentation. <https://platform.openai.com/docs>
6. Ministry of ICT, Bangladesh. (2022). Digital Bangladesh Vision 2041. Government of Bangladesh.
7. a2i Programme. (2023). Access to Information — Digital Literacy Initiative. Government of Bangladesh. <https://a2i.gov.bd>
8. SSLCommerz. (2024). Payment Gateway Integration Guide. <https://sslcommerz.com>
9. bKash Limited. (2024). bKash Payment Gateway for Business. <https://pgw.bkash.com>
10. Sommerville, I. (2016). Software Engineering (10th ed.). Pearson Education. [SRS and Requirements Engineering methodology]